Options

1) Forward Contract

S = Stock Price
K = Strike Price

Payoff = $S_T - K$ (call)
$K - S_T$ (Put)

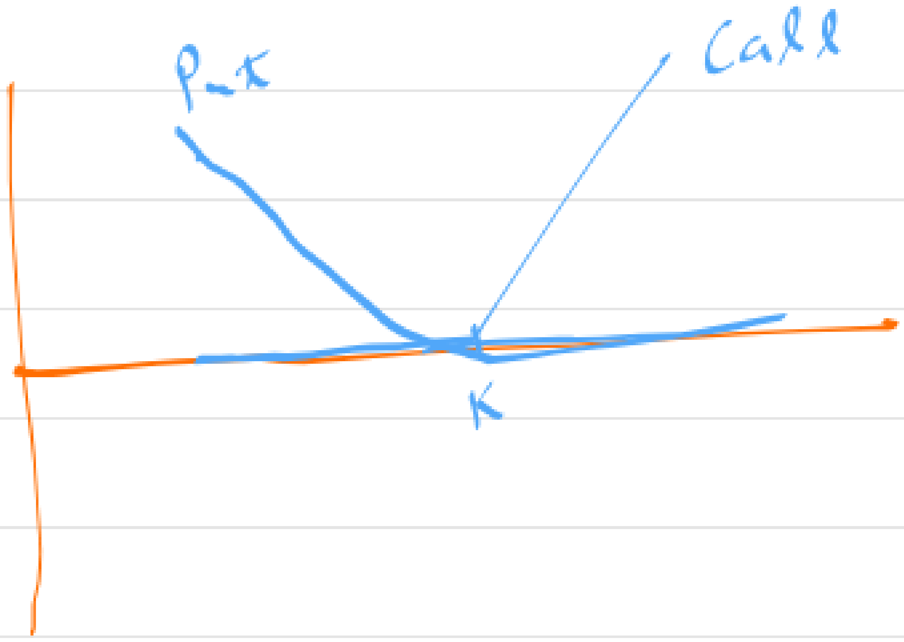
T = 1 year

$\boxed{S_T \to K}$

2) European Option

Payoff ⇒ Call ⇒ $\max(0, S_T - K)$
Put ⇒ $\max(0, K - S_T)$

⇒ Option, exercise at time of year maturity.



3) American Option

=> Exercise => Any time $t \le T$.

=> Binomial model

=> Call (Right to buy)

$$Ke^{-rT} \longleftarrow K$$

Put (Right to sell)

$$K \longrightarrow Ke^{r(T-t)}$$

=) Monte - Carlo

$$S_{T+dt} = S_t \, e^{\left((r-\frac{1}{2}\sigma^2)\,dt + \sigma\sqrt{dt}\,Z\right)}$$

$$\ln S_{T+\Delta t} = \ln S + \underbrace{\boxed{\left(r-\frac{1}{2}\sigma^2\right)\Delta t}}_{nudt} + \sigma\sqrt{\Delta t}\,Z$$

$$\Delta \ln S =$$

$$\begin{array}{ccc} & & \ln S_0 \quad \ln S_0 \quad \ln S_0 \quad \ln S_0 \quad \ln S_0 \\ S_1 & \ln S_1 & \\ S_2 & \ln S_2 & \left[\begin{array}{ccccc} \Delta \ln S & \vdots & \vdots & \vdots & \vdots \\ \Delta \ln S & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{array}\right] \\ S_3 & \ln S_3 & \\ \vdots & \vdots & \\ S_N & \ln S_N & \end{array} \quad N \times M$$

Z (−1 to 1)

$(S_1 S_2 S_7)^{1/3}$

N => Time Steps

M => No. of Sim.

$\max(S-K, 0)$ => avg Call

$\max(K-S, 0)$ => avg Put

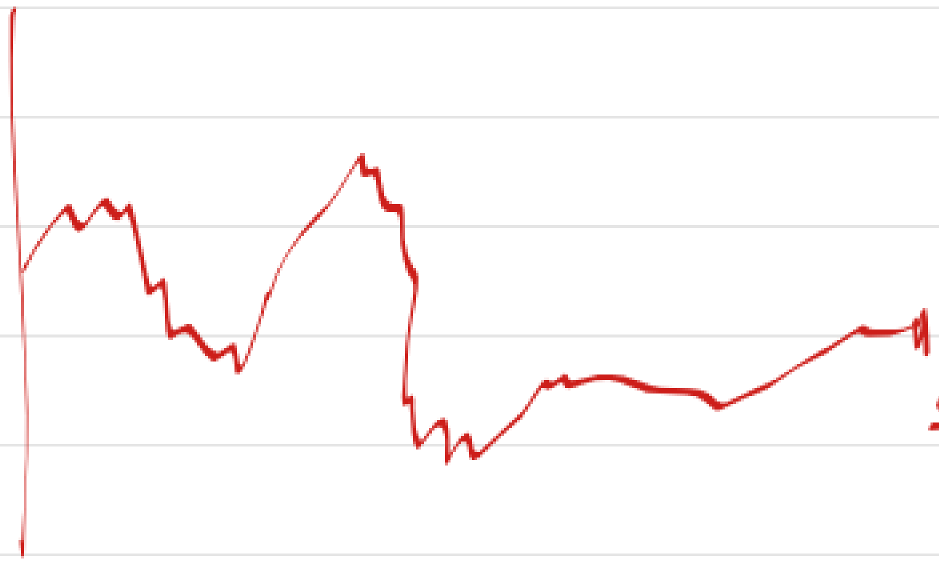
Matrix => numpy $\boxed{(C)}$ → fast

$N \approx 500$ (252)

$M \approx 10^5$

4) Asian Options

Time of Maturity (T)



$= \boxed{A}$

S_T Avg$(A - K, 0)$

Avg$(K - A, 0)$

Average → Arithmetic → Monte-Carlo

↳ Geometric → Monte-Carlo

↳ Black Scholes

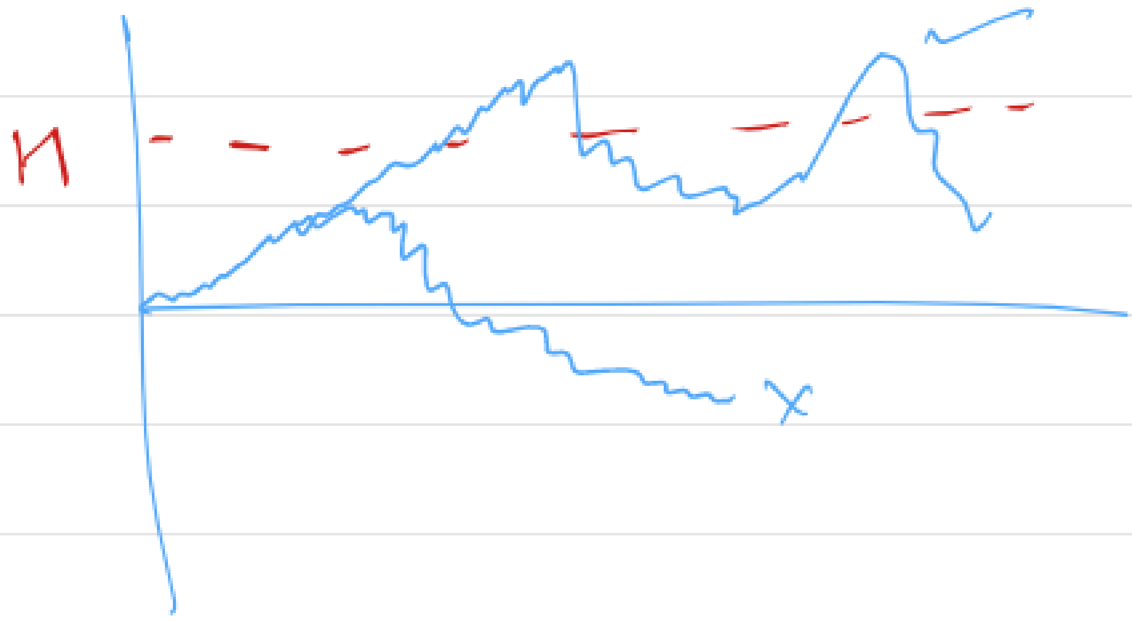
$$C_{\text{geometric}} = e^{-rT}\left[e^{\hat{\mu}+\frac{\hat{\sigma}^2}{2}} N(d_1) - K N(d_2)\right]$$

where:

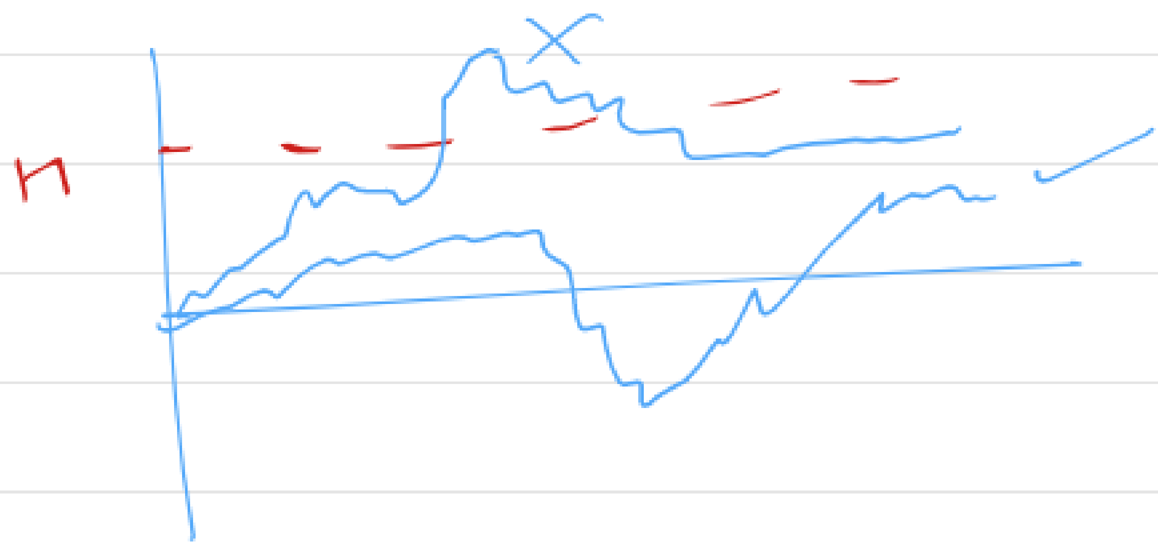
- S_0: current stock price
- K: strike price
- r: risk-free rate
- T: time to maturity
- σ: volatility
- q: dividend yield
- $\hat{\mu} = \ln(S_0) + (r - q - \sigma^2/2)\frac{T}{2}$: adjusted mean for the average
- $\hat{\sigma}^2 = \frac{\sigma^2 T}{3}$: adjusted variance for the average
- $d_1 = \frac{\hat{\mu}+\hat{\sigma}^2-\ln(K)}{\hat{\sigma}}$: standardized distance measure
- $d_2 = d_1 - \hat{\sigma}$: risk-adjusted distance measure
- $N(\cdot)$: cumulative standard normal distribution

5) Barrier Options

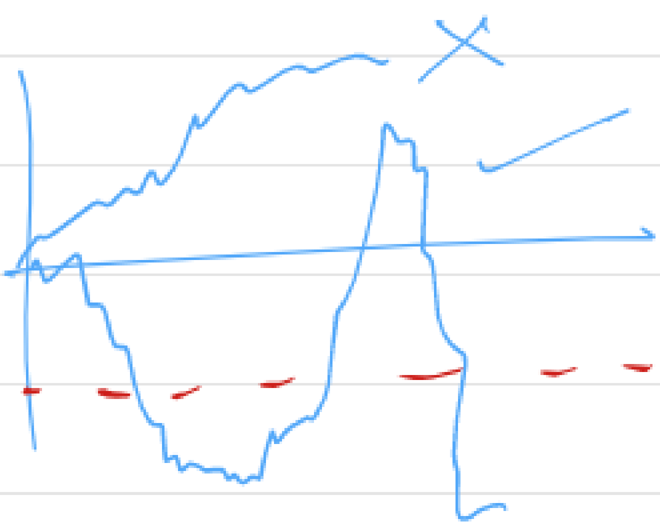
1) up and Knock in



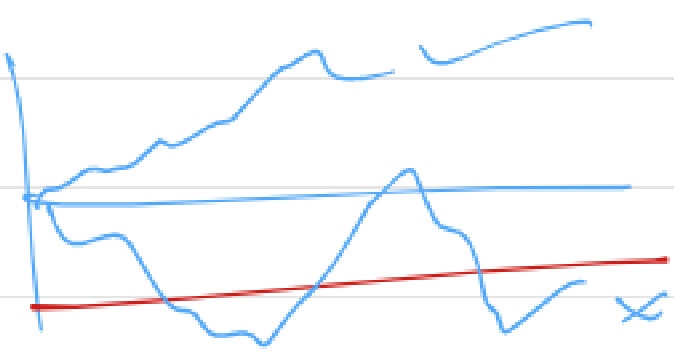
2) up and Knock out

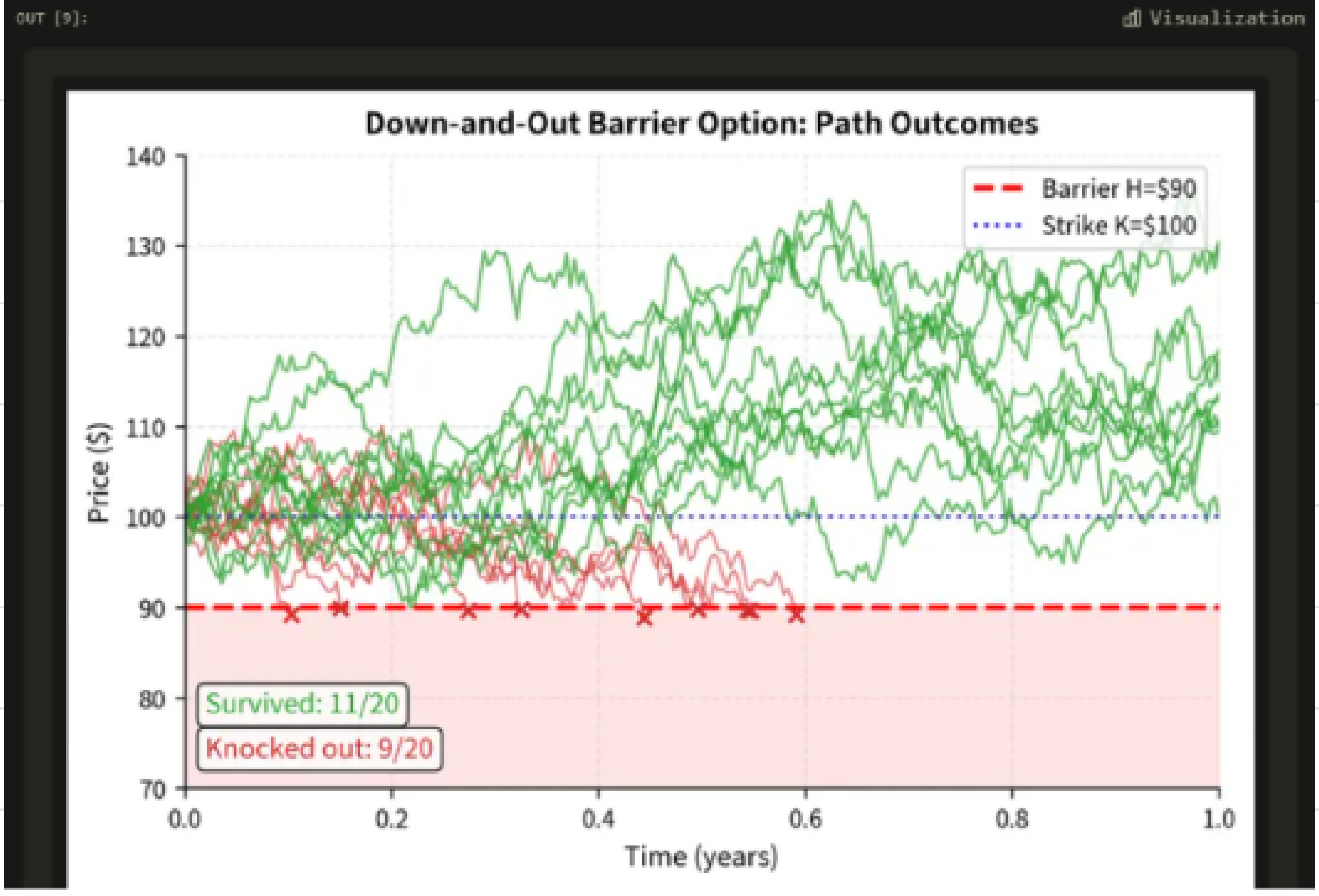


3) down and knock in



4) down and knock out





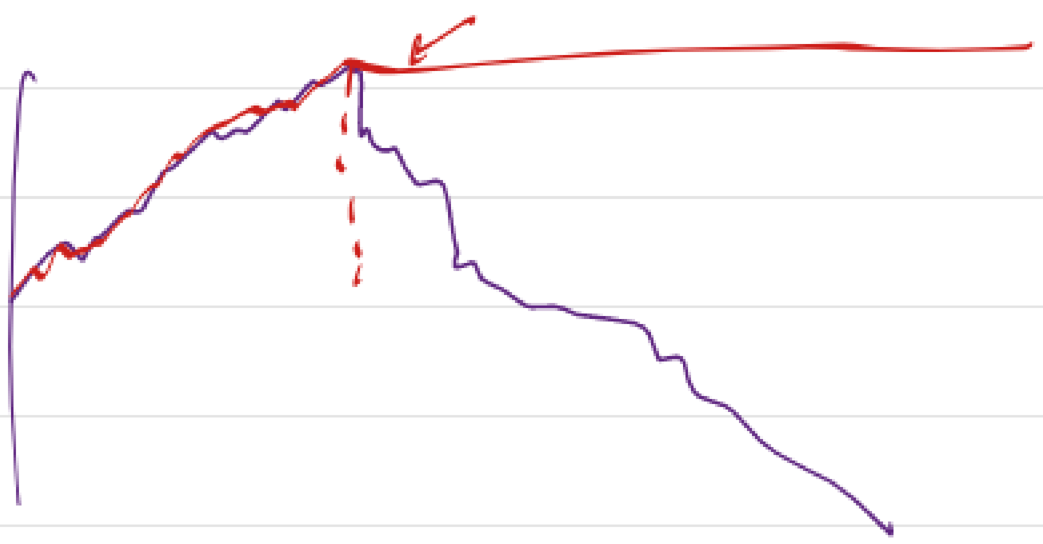
6) Lookback Options

1) Floating

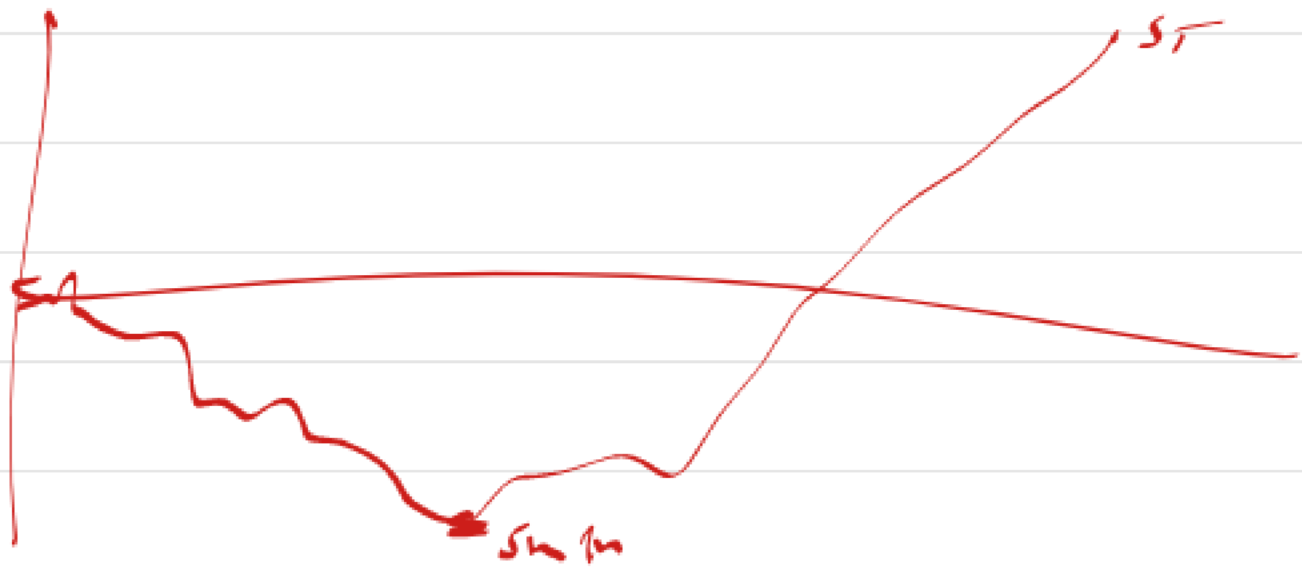
$$(S_T - S_{min})$$

2) Fixed*

$$(S_{max} - K, 0)$$



Fixed



Floating